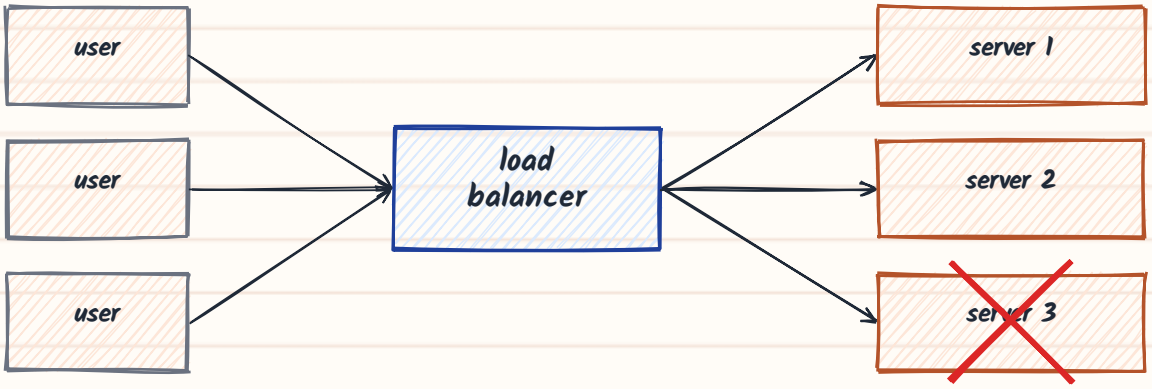
Load Balancers

Lesson 7 · L4 vs L7, algorithms, health checks & HA

In this lesson → ① Many servers ② What an LB does ③ L4 vs L7 ④ Algorithms ⑤ Health checks ⑥ LB as SPOF ⑦ Real-world choices ⑧ Interview Q&A

Running project: SnapLink, our photo-sharing app that grows from 1 server to 100M users.

1 What a load balancer does



health check fails → stop sending traffic

- ✓ Spreads requests across servers
- ✓ Health checks: skips broken servers
- ✓ Lets you add/remove servers with no downtime
- ✓ Often: TLS termination, compression, rate limiting

2 L4 vs L7

L4 (transport)

sees: IP + port (TCP / UDP)

very fast · simple
can't read HTTP

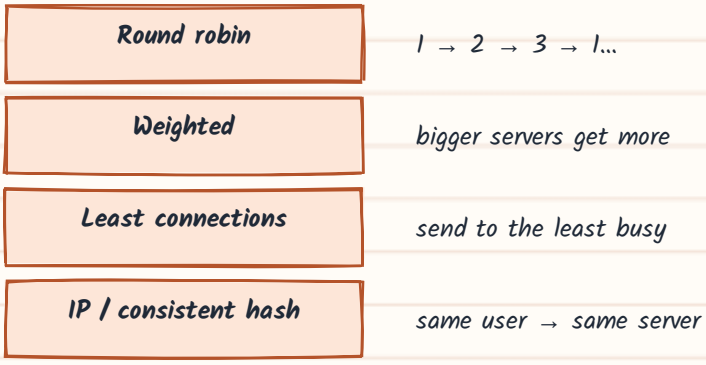
L7 (application)

sees: URL, headers, cookies

/api → API servers
/img → image servers

	L4 (transport)	L7 (application)
Sees	IP, port (TCP/UDP)	URL, headers, cookies, body
Speed	very fast	a bit slower
Can do	pass-through, TCP/UDP	path/host routing, TLS termination, auth, rate limits
Examples	AWS NLB, K8s Service	Nginx, Envoy, AWS ALB, Ingress

3 Balancing algorithms



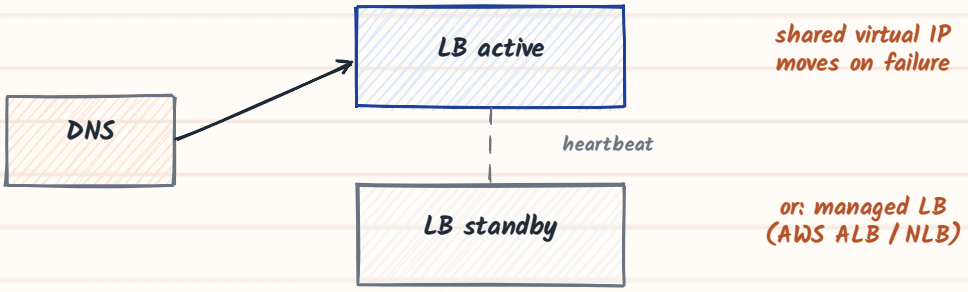
Algorithm	Best when
Round robin	servers equal, requests similar
Weighted round robin	servers of different sizes
Least connections / least response time	request durations vary (uploads, long queries)
IP hash / consistent hashing	same client → same server (caches, sessions)

4 Health checks

Type	How	Note
Active	LB calls /health every N s	remove after K failures
Passive	watch real responses	errors / timeouts → eject

> **Tip**
> Keep /health cheap; don't fail it because one dependency is slow, or every server drops out at once.

5 The LB must not be a SPOF



→ Active-passive pair + floating (virtual) IP
→ Managed cloud LBs are redundant by design
→ DNS across multiple regional LBs

6 Real-world choices

Where	L4	L7
Software	HAProxy (TCP)	Nginx, HAProxy, Envoy
AWS	Network Load Balancer	Application Load Balancer
Kubernetes	Service	Ingress / Gateway API

SnapLink
DNS → regional **L7** LB (TLS termination) → /api to API servers, /img to image servers.

7 Sticky sessions

→ LB pins a user to one server (cookie / IP hash)
→ Uneven load; session lost on failure
→ Prefer stateless servers (SD-03)

8 Interview Corner

Q. What does a load balancer do?

A. Distributes traffic across healthy servers and removes unhealthy ones via health checks.

Q. L4 vs L7 load balancer?

A. L4 routes by IP/port; L7 understands HTTP and routes by path, host, headers or cookies.

Q. Which algorithm when request times vary?

A. Least connections (or least response time).

Q. How to keep a user on the same server?

A. IP hash / consistent hashing or sticky cookies (but prefer stateless).

Q. How do you avoid the LB being a SPOF?

A. Redundant active-passive pair with a virtual IP, or a managed LB, plus DNS across regions.

⚡ Revise in 60 seconds

- → LB = spread + hide failures
- → **L4**: IP/port, fast
- → **L7**: HTTP-aware, smart
- → Health checks: active/passive
- → RR · weighted · least conn · hash
- → Redundant LB pair / managed
- → TLS termination at L7
- → Prefer stateless servers

Next lesson → SD-08
CDN & edge caching.
Watch on YouTube: **@TheCodeNotebook** · Notes: **thecodenotebook.com**